THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The SI unit of power is the: 1

(a) watt **(b)** kg **(c)** joule **(d)** newton

2. In a lever of the first class, the fulcrum is: 1

(a) absent **(b)** at the end next to effort

(c) between the load and the effort **(d)** at the end next to load

3. Work done is negative when the force acts: 1

(a) never **(b)** perpendicular to the displacement

(c) along the displacement **(d)** opposite to the displacement

4. 1 kilowatt hour (kWh) equals: 1

(a) 3.6 x 10^3 J **(b)** 3.6 x 10^6 J **(c)** 3600 J **(d)** 1000 J

5. A wheelbarrow is a lever of the: 1

(a) third class **(b)** second class **(c)** first class **(d)** fourth class

6. The energy possessed by a body due to its motion is: 1

(a) chemical energy **(b)** heat **(c)** kinetic energy **(d)** potential energy

7. A machine does 500 J of work in 10 s. Its power is: 1

(a) 510 W **(b)** 0.02 W **(c)** 50 W **(d)** 5000 W

8. Tweezers are a lever of the: 1

(a) first class **(b)** second class **(c)** third class **(d)** none

9. The energy possessed by a body due to its position or state is: 1

(a) sound **(b)** kinetic energy **(c)** light **(d)** potential energy

10. A body of mass 2 kg moves at 3 m/s. Its kinetic energy is: 1

(a) 3 J **(b)** 18 J **(c)** 9 J **(d)** 6 J

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State the law of conservation of energy. 2

12. Classify the levers: scissors, wheelbarrow and tweezers. 2

13. Calculate the kinetic energy of a 0.2 kg ball moving at 10 m/s. 2

14. Why is no work done when a person holds a bag at rest? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Derive the expression for gravitational potential energy. 3

16. Explain how a lever helps us. Define load arm and effort arm. 3

17. Name the energy changes in: 3

(a) a dynamo

(b) a loudspeaker

(c) a burning candle.

OR

A child slides down a smooth slide of height 2.5 m. Find the speed at the bottom (g = 10). 3

18. State the principle of the lever and the condition of balance. 3

OR

A car of mass 1000 kg moving at 20 m/s is stopped by brakes. Find the work done by the brakes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A body of mass 4 kg is thrown vertically upward with a speed of 20 m/s. Find its KE and PE at the start, at 10 m height and at the top (g = 10). 5

OR

Explain the different forms of energy with examples and conversions: a hydroelectric plant, a car engine and a solar cell. 5

20. Explain the three classes of levers with examples and diagrams in words. 5

OR

Differentiate between work and power with units and examples, and calculate the power of a man who does 3600 J of work in 2 minutes. 5

END OF QUESTION PAPER — Total: 40 Marks

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